VIT®

Vellore Institute of Technology

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22BAI1266

AFZAL

Continuous Assessment Test I - March 2023

Programme	: B.Tech.	Semester	: Winter Semester I year 2022-23
Course	: Differential Equations and Transforms	Code	: BMAT102L
		Slot	: D1+TD1+TDD1
Faculty	: Dr. Radha, Dr. Nathiya, Dr. Harshavarthini, Dr. David Raj, Dr. Ashish Kumar, Dr. Karan Kumar Pradhan, Dr. Manoj Kumar Singh, Dr. Prosenjit Paul, Dr. Abhishek Kumar Singh, Dr. Sandip Saha, Dr. Revathi, Dr. Vijay Kumar, Dr. Berin Greeni	Class ID	: CH2022232300591, 594, 595, 622, 627, 629, 630, 628, 593, 631, 590, 634, 680, 592
Time	: 90 Minutes	Max.Marks	: 50

Answer ALL the questions ($5 \times 10 = 50$ marks)

1. Find the singular integral of $(p+q)(z-px-qy)=1$. [10]

2. (a) Find the complete solution of $q = py + p^2$. [3]

   (b) Solve the following PDE using separation of variables method [7]
   $\frac{\partial z}{\partial x} + 4z = \frac{\partial z}{\partial t}$, given that $z(x,0) = 4e^{-3x}$.

3. Find the Laplace transformation of the following:

   (a) $f(t) = e^{3t}(\cosh t + \sin^2 t)$ [5]

   (b) $f(t) = \sqrt{t} - \frac{(\sqrt{t})^3}{3!} + \frac{(\sqrt{t})^5}{5!}$ [5]

4. Solve the following differential equation [10]
   $x^2y'' + 5xy' + 4y = x\log x$.

5. (a) Solve the following differential equation by the method of undetermined coefficients [5]
   $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = e^x$.

   (b) An electric circuit consists of an inductance $L = 2H$, a resistance $R = 14\Omega$ and a condenser of capacitance of $C = 0.05F$. Find the charge $q(t)$ and current $i(t)$ at any time t. Given that $q(0) = 1$ and $i(0) = 4$. [5]